

Zhiheng Zhang

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github.com/zhiheng-zhang-Mera

Melbourne, Australia

Research Profile

Research engineer studying reliable long-horizon coding agents, execution-grounded replanning, persistent state and outcome-based validation.

Education

University of Melbourne , Australia	2025–2026 (expected)
Master of Computer Science (research project underway)	
University of British Columbia , Canada	Completed 2024
Bachelor of Science in Computer Science	

Selected Research Engineering Projects

DS-Hns (Completed, Frozen) Predecessor: durable agent execution

github.com/zhiheng-zhang-Mera/DS-Hns

- Completed and froze predecessor DS-Hns with durable repository-work lifecycles, scheduling, restart/resume recovery and isolated extensions.
- Used explicit terminal-state and recovery evidence to study the gap between continued process execution and genuine task completion.

Codex Boss (Completed, Frozen) Predecessor: agent governance

github.com/zhiheng-zhang-Mera/Codex-Boss

- Completed and froze predecessor Codex Boss with agent orchestration, critique and evidence-governed acceptance foundations.
- Preserved unresolved negative cases and traceable failures rather than treating superficial success signals as proof of completion.

Utopia / Digital City (Completed, Frozen) Integrated successor

github.com/zhiheng-zhang-Mera/utopia

- Completed and froze Utopia as the integrated successor to DS-Hns and Codex Boss; delivered Web/Android control, authenticated routing and persistent task-state contracts.
- Verified a three-end setup with two physical Windows execution workers and Android control; PCF integration includes bounded real cross-host task results and explicit resource-observation contracts.
- Maintained research-evaluation evidence for fault injection, replay, traceable results and independent bounded reproducibility; wearable deployment and full dual-server failover remain future work.

Research Methods and Skills

System prototyping; multi-device software integration; long-running task lifecycles; fault injection; reproducible experiment and evidence design; test and failure analysis; independent cross-host review.

Proposed Doctoral Research

How an agent can revise a failing repository-scale plan without invalidating previously accepted work or drifting away from the user's actual acceptance criteria. Experimental priority: Use multi-step repository tasks with seeded CI/build failures, alternative planning decisions and verification receipts; compare state drift, false-completion decisions, genuine patch validation and human intervention.

Future project: Celestial is the planned redesign and next iteration of the completed, frozen Utopia; it is not yet a completed product.